

Program : Diploma in Instrumentation Engineering	
Course Code : 5083B	Course Title: Laser and Optical Instrumentation
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week : 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge about the fundamentals of optics and opto-electronic devices.
- To illustrate the applications of Lasers in various fields.
- To understand about fiber optic communication.
- To illustrate the principles of transducers in fiber optic sensors.

Course Prerequisites:

Topic	Course code	Course name	Semester
Lasers		Applied physics II	2
Fiber optic sensor		Sensors and transducers	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Describe the properties of light and fundamentals of Laser	16	Understanding
CO2	Illustrate the classification and applications of Laser	14	Applying
CO3	Explain the fundamentals of fiber optic communication systems and optoelectronic components	15	Understanding
CO4	Illustrate the principles of transducers in fiber optic sensors	13	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the properties of light and fundamentals of Laser		
M1.01	Explain the Phenomenon and laws of reflection, refraction, dispersion, interference, diffraction and polarization	6	Understanding
M1.02	Describe Young's double slit experiment for interference phenomenon	3	Understanding
M1.03	Illustrate Newton's ring experiment and its application	3	Understanding
M1.04	Explain the characteristics and principles of Laser	4	Understanding
Contents: Phenomenon and laws of reflection - refraction - dispersion-interference - Young's double slit experiment - phase difference and path difference - constructive and destructive interference - conditions for bright and dark fringes - conditions for distribution of light intensities - Newton's Rings - Application of Newton's rings - diffraction - difference between interference and diffraction - polarization - Characteristics of laser beam - Interaction between matter and radiation - absorption - spontaneous emission - stimulated emission - Thermal equilibrium - Population inversion - Basic requirements for producing laser			
CO2	Illustrate the classification and applications of Laser		
M2.01	Explain the Construction and working of Ruby laser, Nd-YAG laser, He-Ne Laser, Argon Ion Laser, CO ₂ Laser, Liquid dye Laser, Semiconductor Laser	8	Understanding
M2.02	Identify the industrial, medical, scientific, military and miscellaneous applications of laser	5	Applying

M2.03	Identify the application of Laser in instrumentation area	1	Applying
	Series Test – I	1	
Contents: Classification of laser - Construction and Working of - Ruby laser - Nd - YAG laser - He - Ne Laser - Argon Ion Laser - CO2 Laser - Liquid dye Laser - Semiconductor Laser- Industrial Applications of laser (laser welding, cutting, drilling) - Medical applications of laser - Scientific applications of laser - Military Applications of laser - Miscellaneous applications of laser(laser printer optical disc recording and reading, holography, construction of hologram and reconstruction of image) - Laser application in Instrumentation (displacement measurement - Laser Doppler velocity meter)			
CO3	Explain the fundamentals of fiber optic communication systems and optoelectronic components		
M3.01	Describe the construction and principle of operation of optical fiber	2	Understanding
M3.02	Explain the modes of operation of optical fiber	2	Understanding
M3.03	Explain the block diagram and advantages of fiber optic communication	4	Understanding
M3.04	Describe the construction and working of optoelectronic components(photo diode- PiN diode- Avalanche photo Diode-photo transistor- LED)	7	Understanding
Contents: Principle of Optical Fibers - critical angle - total internal reflection - optical fiber construction - acceptance angle and Numerical aperture - modes of operation of optical fibers - Single mode step index fiber - multi mode step index fiber - multimode graded index fiber - block diagram of Fiber optic Communication System - Advantages of fiber optic communication - Construction and working of photo diode - PiN diode - Avalanche Photo Diode-photo transistor - LED - principle of operation - LED drive circuits.			
CO4	Illustrate the principles of transducers in fiber optic sensors		
M4.01	Describe the general structure of fiber optic sensor.	2	Understanding
M4.02	Differentiate between passive and active optical fiber sensors	1	Understanding
M4.03	Application of fiber optic sensor (displacement, pressure, temperature, force, torque, strain, level, flow and humidity measurement)	8	Applying

M4.04	Describe the advantages of fiber optic sensor	2	Understanding
	Series Test – II	1	
Contents: General structure of fiber optic sensor-passive and active optical fiber sensors- Displacement sensor- Pressure sensor- Temperature sensor- Force sensor- Torque sensor- Strain sensor - Level sensor- Flow sensor-humidity sensor-Advantages of fiber optic sensor.			

Text / Reference:

T/R	Book Title/Author
R ₁	S. Nagabhushana and N. Sathyanarayana, Lasers and Optical Instrumentation, I K International Publishing House
R ₂	N. Subrahmanyam & Brijlal, A Text Book of Optics, S Chand publications
R ₃	B.L. Theraja, Engineering Physics, S Chand publications
R ₄	Keiser, Optical Fiber Communication System, TMH
R ₅	K. Thyagaraja & Ghatak, Laser theory & application, Springer
R ₆	T.L Singal, Optical fiber communications, Cambridge university press

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=yfawFJCRDSE
2	https://www.youtube.com/watch?v=gIGOXNlvMsg
3	https://www.youtube.com/watch?v=Z2Kme6ozj7U
4	https://www.youtube.com/watch?v=Gsj44ObhL24
5	http://www.powershow.com/view/348ef-
6	https://www.docslides.com/cheryl-pisano/fiber-optical-sensor
7	http://www.circuitstoday.com/how-a-led-works-light-emitting-diode-working
8	https://en.wikipedia.org/wiki/List_of_laser_applications